7. Time Does Not Exist

> It must not be claimed that anyone can sense time by itself apart from the movement of things.
>
> – Lucretius, *De rerum natura*[1]

The alert reader will have realized that in the preceding chapter little attention was given to time. And yet Einstein showed, over a century ago, that we cannot separate time and space, that we must think of them together as a single whole: spacetime. The moment has come to rectify this and bring time back into the picture.

Research on quantum gravity has revolved for years around spatial equations, before having the courage to confront time. In the last fifteen years, a way of thinking about time has begun to emerge. I'll try to explain it.

Space as an amorphous container of things disappears from physics with quantum gravity. Things (the quanta) do not inhabit space, they dwell one over the other, and space is the fabric of their neighbouring relations. As we abandon the idea of space as an inert container, similarly, we must abandon the idea of time as an inert flow along which reality unfurls. Just as the idea of the space continuum containing things disappears, so, too, does the idea of a flowing continuum 'time' during the course of which phenomena happen.

In a certain sense, space no longer exists in fundamental theory; the quanta of the gravitational field are not *in* space. In the same sense, time no longer exists in the fundamental theory: the quanta of gravity do not evolve *in* time. Time just counts their interactions. As evidenced with the Wheeler–De Witt equation, the fundamental equations no longer contain the time variable. Time emerges, like space, from the quantum gravitational field.

This was already partially true for classical general relativity, where time already appears as an aspect of the gravitational field. But as long as we neglect quantum theory, we can still think of spacetime in a rather conventional manner, like the tapestry in which the story of the rest of reality unfolds, even if it is a dynamical, moving tapestry. The moment we take quantum mechanics into account, we recognize that time, too, must have those aspects of probabilistic indeterminacy, granularity and relationality which are common to all of reality. It becomes a 'time' markedly different from all that we have hitherto meant by the word.

This second conceptual consequence of the theory of quantum gravity is more extreme even than the vanishing of space.